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Facial Emotion Recognition with Dimensionality Reduction: A Comparative Study of PCA, FA, and Combined PCA-FA with a CNN Baseline

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Abstract. Facial emotion recognition (FER) is used in health, education, and human-computer interaction, yet accuracy suffers on imbalanced data and deep models can be costly to train. We test whether statistical dimensionality reduction improves efficiency and stability for classical models, and provide a comparative baseline against a CNN. Using FER-2013 (~36k 48×48 grayscale faces; seven emotions: angry, disgust, fear, happy, sad, surprise, neutral), we used an 80/20 stratified hold-out with standard image augmentation (rotation, shift, shear/zoom, flip). For classical models, we flattened images and reduced them with Principal Component Analysis (PCA) and Factor Analysis (FA). With 50 components, PCA compressed 2,304 pixels to 50 while retaining 86.50% variance, shortening classical training without accuracy loss. We trained Support Vector Machines, Random Forests, and Logistic Regression, and compared them with a Convolutional Neural Network (CNN). Metrics included accuracy, precision, recall, and class-wise F1. CNN reached 60.30% validation accuracy. Classical models: SVM ~43–44%, Random Forest ~43–45% (overfitting), Logistic Regression ~37%. Among classical pipelines, combined PCA-FA yielded the strongest gains (macro-F1 0.41, weighted-F1 0.42). Per-class F1 showed the bottleneck: happy was recognized best (CNN F1 ≈ 0.26), while disgust was rarely detected (≈ 0.00). We intentionally avoided class weighting to preserve a clean baseline. This hybrid comparison shows how PCA and FA shape FER performance for classical models and where a standard dataset falls short. Dimensionality reduction cuts redundancy and improves efficiency for classical models, but class imbalance dominates errors. The study points to next steps: targeted augmentation and class weighting for rare emotions, larger and cleaner data, and transfer learning for subtle classes relevant for FER on resource-constrained devices in mobile health, classrooms, and on-device surveillance.

Keywords: Facial emotion recognition, Convolutional neural networks, Principal component analysis, Factor analysis, Dimensionality reduction.

Heavy-Ion Induced Nuclear Reactions for Hadron Therapy and Medical Applications

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Abstract: Heavy-ion induced nuclear reactions are of increasing interest in both fundamental nuclear physics and applied medical research, particularly in hadron therapy and medical radioisotope production. In this work, detailed analyses of heavy-ion reactions were performed using the Particle and Heavy Ion Transport code System (PHITS) with JAERI Quantum Molecular Dynamics (JQMD) and Intranuclear Cascade Liège (INCL) models. Double differential cross sections (DDXs) of neutrons, protons, and light ions were calculated and benchmarked against experimental data to assess model performance. The study shows that both JQMD and INCL reproduce the overall features of neutron and proton energy spectra, with deviations typically within 15–30% at forward angles. Discrepancies increase at backward angles and for light-cluster emission (d, t, α). INCL performs better at low to intermediate energies, while JQMD is more accurate for neutron spectra. However, neither model adequately describes light-ion emission. Overall, the outcomes confirm that PHITS, utilizing JQMD and INCL models, is a useful tool for simulating nuclear reactions, contributing to the optimization of heavy-ion therapy, treatment planning, and secondary radiation evaluation.

Keywords: Heavy-ion reactions, Double-differential cross-section (DDX), PHITS, Fragmentation models, Ion beam therapy.